Results

For this Ben Olschar, 108021211678 and Frederik Hüttemann, 108021215247 cooperated with eachother. We did implement the code on our own, but agreed to using one version. The discussion and results are made in cooperation.

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Before starting, we need to get some information about the GPU (RTX
4060ti: 8.9 Compute Capability) launch configuration:
Number of SMs: 34
Max threads per SM: 1536
Max blocks per SM: 24
Thread warp size: 32
Suggested BLOCK_SIZE: 64
Suggested GRID_SIZE: 816
We also need to know the number of elements in our 40 MB float array:
For 40 MB array size, use BLOCK_SIZE of 256 and GRID_SIZE of 544 for
optimal performance.
A 40 MB array contains 10485760 float elements.


Running Task 1: CPU Reduction:
CPU Reduction Time: 0.022748 sec
Running Task 2: GPU Reduction Test with atomic add on global memory:
Results match.
GPU Reduction Time: 0.014661 sec
Running Task 3: GPU Reduction Test with cascaded reduction:
GPU Cascaded Reduction Time: 0.000234 sec
Results match.
```

PROF

GPU Configuration:

Parameter	**Value**
Number of SMs	34
Max threads per SM	1536
Max blocks per SM	24
Warp size	32
Suggested BLOCK_SIZE	64
Suggested GRID_SIZE	816
Array size	40 MB

Parameter	Value
Number of float elements	10,485,760
Optimized BLOCK_SIZE (40 MB)	256
Optimized GRID_SIZE (40 MB)	544

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Task	Description	Time (sec)	Status
Task 1	CPU Reduction	0.022748	–
Task 2	GPU Reduction (atomic add, global memory)	0.014661	Results match
Task 3	GPU Cascaded Reduction	0.000234	Results match